WeBoosta

Visible to Agents, Not Just Humans

Transforming fragmented, manual marketing into a unified, AI-driven pipeline

UM Hackathon 2026

Domain 1: AI Systems &
Agentic Workflow Automation

Team UMami

The Paradigm Shift

The New Reality

Travelers don't just "Google" anymore. They use AI agents to plan, select, and book

The Upcoming Future

"Marketing to AI" is the new SEO (AEO - AI Engine Optimization)

The Potential Threat

If an AI can’t easily parse your hotel's data, you are invisible

The Human Limit

Old Way: Fragmented & Manual

Data Scramble
Marketing teams manually scour Agoda, read reviews, and guess what algorithms want

Human Bottleneck
Reading unstructured data and inferring patterns is slow, inconsistent, and doesn't scale.

Endless Loop
very update requires manual re-evaluation, re-testing, and re-writing.

Result
Messy text built for human eyes, completely unreadable to the AI agents actually making the decisions

Solution
WeBoosta Way

A Unified Agentic Pipeline

Automated Evaluation
We simulate an AI travel agent to evaluate a property's digital footprint

Zero Guesswork
Google Gemini 3.1 Pro interprets unstructured data and converts it into structured, actionable insights

Machine-Readable Outputs
Generates AI-friendly schemas, clear scoring, and targeted recommendations

Result
Faster, scalable, and mathematically proven optimization.

The Multi-Agent Loop

Evaluate → Optimize → Re-evaluate

Discovery & Extraction

Web crawling (Firecrawl + Tavily) pulls raw, unstructured data

AI Decision Simulator:

Simulates an AI agent to calculate a baseline "Selection Score"

Gap Analyzer & Optimization

Identifies missing trust signals and autonomously generates schema-ready content

Validation (QA)

Verifies factual accuracy—ensuring zero hallucinations

Re-Simulation

Proves the ROI by demonstrating the mathematically improved score

Frontend & 3D Visualization

Core Framework

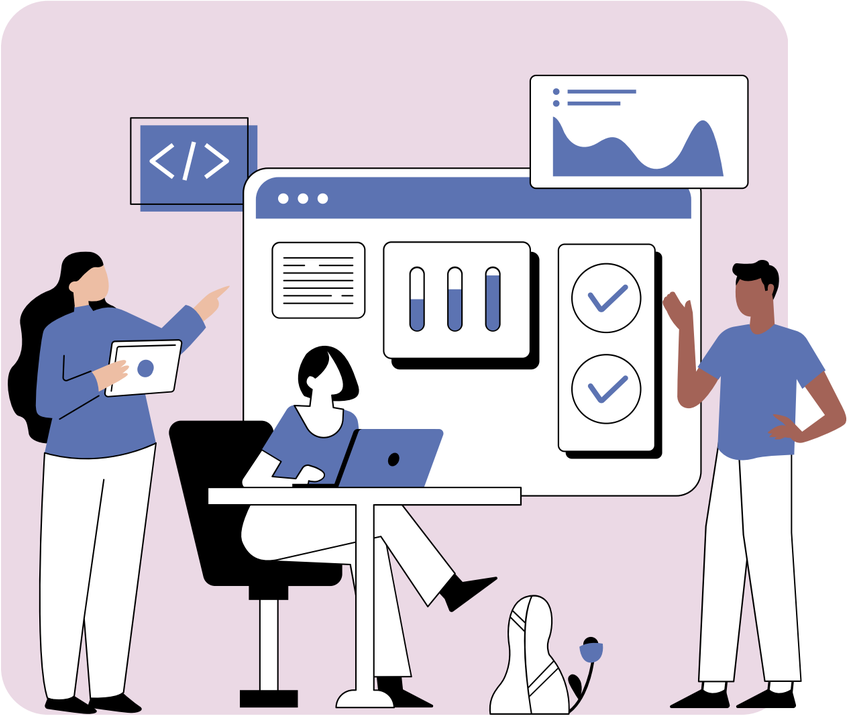
Next.js
Powers our App Router for a lightning-fast, scalable, and responsive web interface

React
Builds our dynamic, component-driven user interface

UI & Motion

Tailwind CSS
Utility-first CSS ensuring a sleek, modern, and highly responsive design system

Framer Motion
Delivers smooth, professional micro-interactions and page transitions



Frontend & 3D Visualization

Data Spatialization

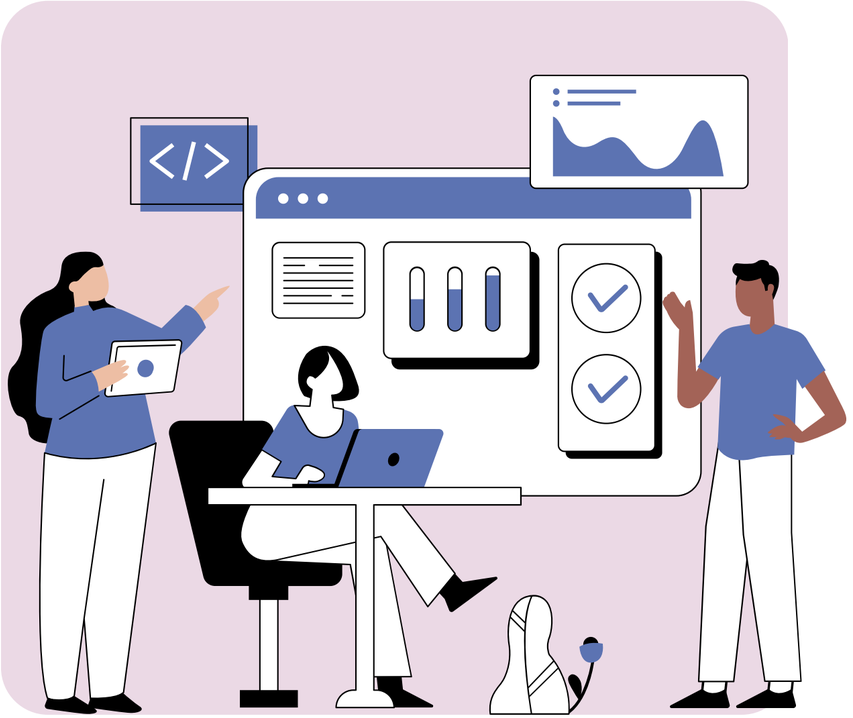
Three.js
Used to create and display animated 3D computer graphics in a web browser

React Three Fiber (R3F)
A powerful React renderer for Three.js that allows us to build the 3D agent simulation

Analytics Rendering

React Query (TanStack Query)
Manages asynchronous state and server-side data fetching

WebSocket
Integrated for real-time updates



Backend & Agentic Orchestration

Language

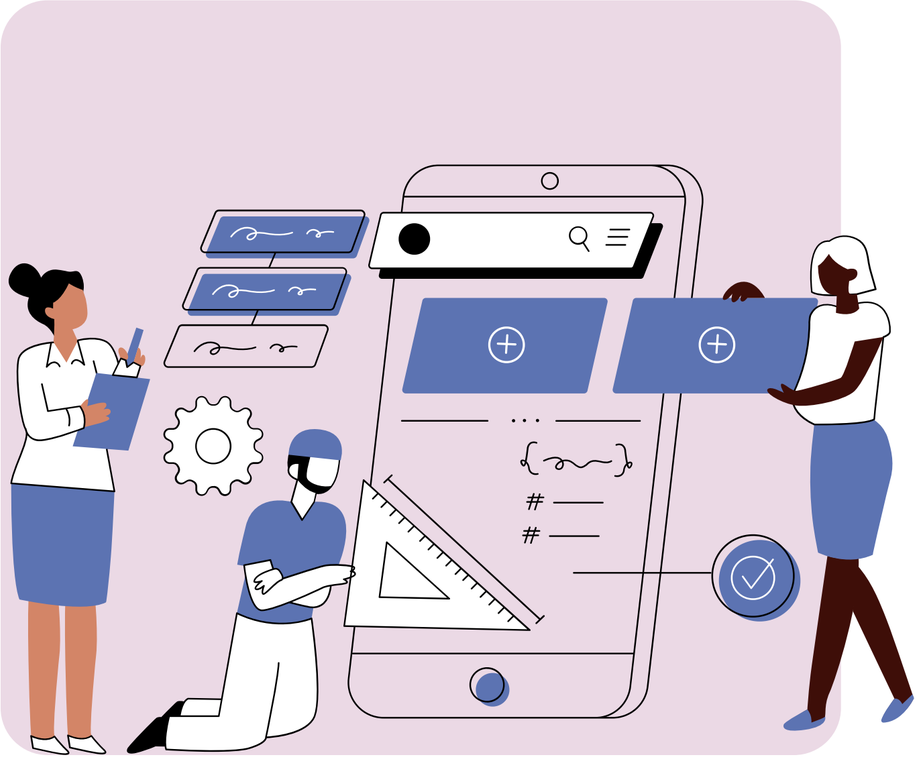
Python
The foundational language of our system, chosen for its robust ecosystem in AI

Framework

FastAPI
High-performance, asynchronous web framework to handle real-time API requests

Orchestration & LLM Integration

LangGraph
The central framework used to chain together complex LLM prompts and manage interactions between our various AI tools



Backend & Agentic Orchestration

Real-Time Communication

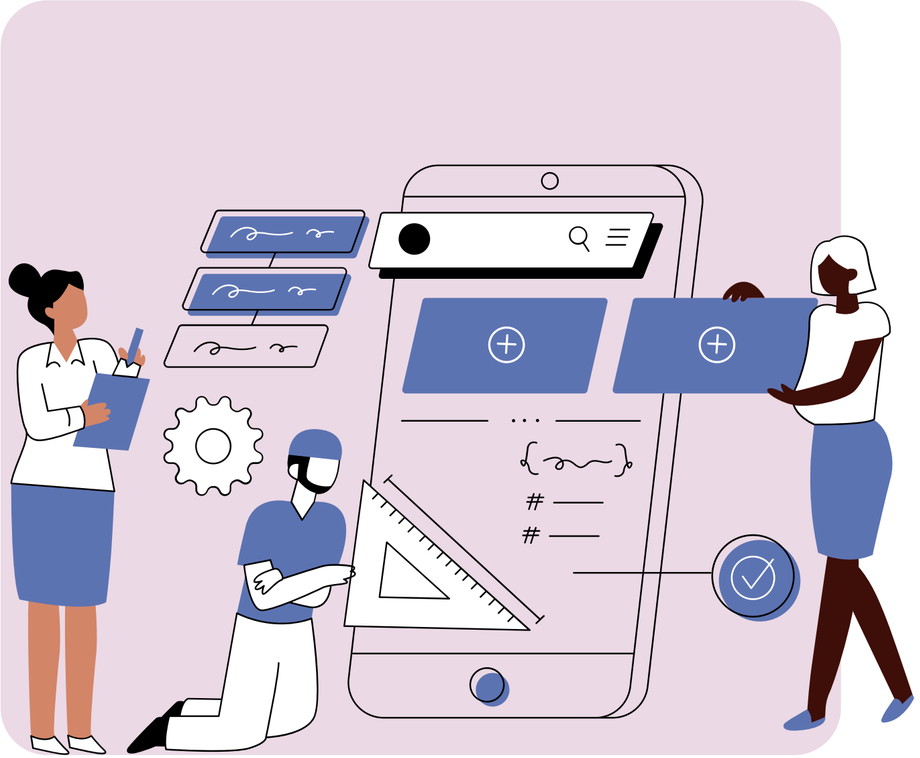
WebSockets
Used by the agent to stream the updates to frontend

Search & Discovery

DuckDuckGo
A privacy-first search tool that allows our agents to perform live web research without tracking or bias

Tavily
Used for deep hotel discovery and market research

Firecrawl
Used for high-fidelity web scraping and content extraction



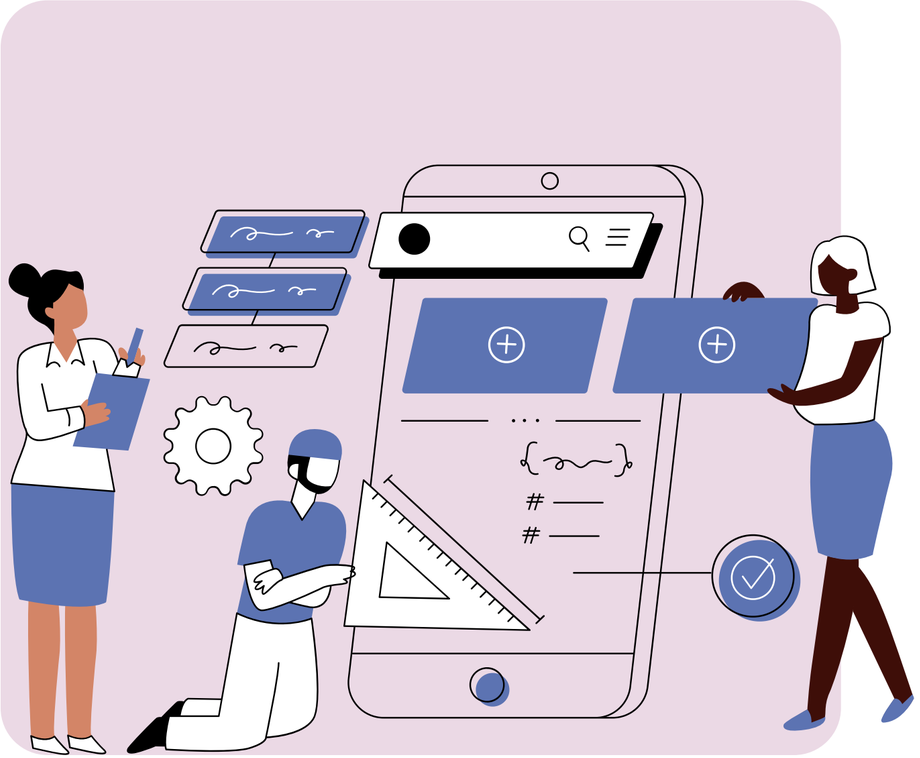
Backend & Agentic Orchestration

Database & Persistence

Supabase
Our primary PostgreSQL database used for long-term storage of hotel profiles, optimization records, and user history

Environment

Pydantic
Used for robust data validation and schema enforcement



WeBoosta Advantage

Iterative
The system improves itself automatically.

Auditable
Human-in-the-Loop (HITL) dashboard ensures final say before saved.

Future-Proof
Moves hospitality marketing out of the search engine era and into the agentic era.

Team Umami

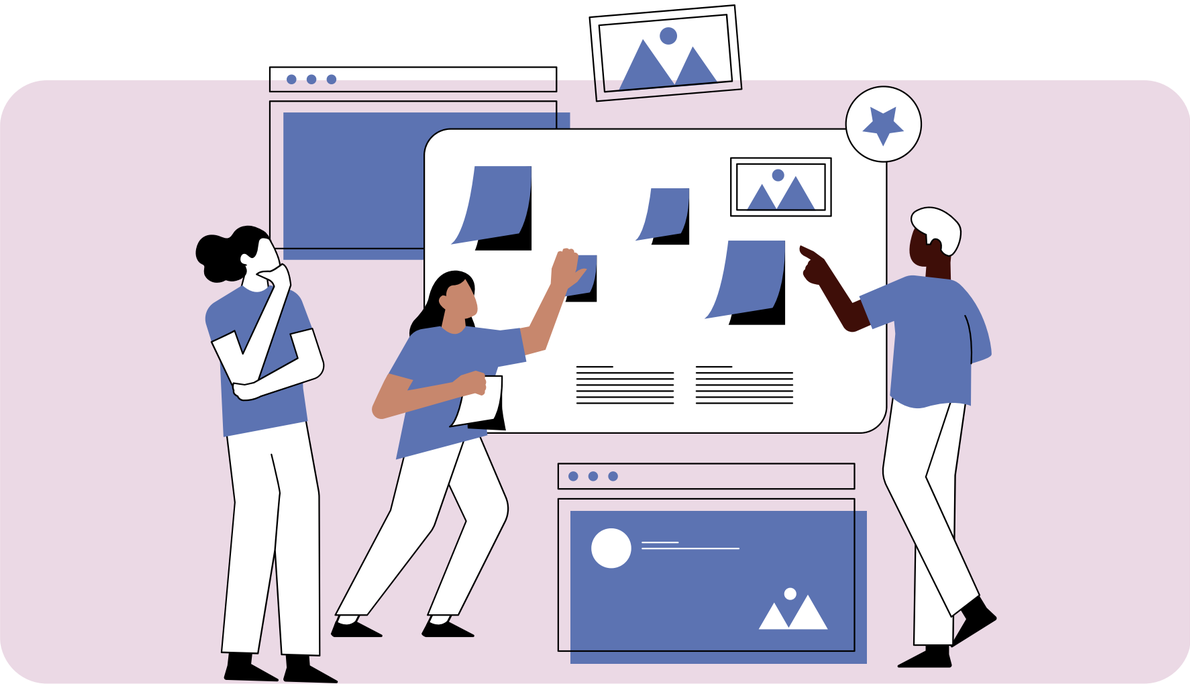
Lee Ping Xian
UI/UX & Workflow Architect (Taylor's University)

Shariq Nauman
Backend Developer (Monash University)

Muhammad Ahmad
QA & Pitch Deck (Monash University)

Francis Paul Narcis
Documentation (Monash University)

Leong Wui Yip
Video Director (Monash University)



THANK YOU SO MUCH

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